

- DSC 550: Week 12
- Final Project - Milestone 1-4
- Martin Hoehler
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Final Project – Predicting a Country’s Banking Z-Score

Introduction

Each country’s financial system depends on the strength of their banks, and the confidence their citizens have in the deposits they place. A country’s ‘Z-Score’ is a measure of the likelihood of default, by comparing “the buffer of a country’s commercial banking system with the volatility of those returns” (World Bank Group, “Metadata Glossary”). Knowing a country’s Z-score can give investors and businesses important information to help them decide the risk of doing business in a country.

Justification

The Z-Score, though, is determined entirely through accounting data, which is “perhaps the most important limitation” (World Bank Group, “Financial Stability”). Being able to demonstrate a relationship between other data and the Z-Score would enable businesses to gauge the safety of a country’s banking system through more than just the tracking of equity and assets.

Further, if a model were made that could predict a Z-Score based on the access citizens have to financial products, it could help financial institutions determine the benefit of outreach to underserved communities.

In the banking industry, a major concern is determining areas of the footprint in which the population is underserved by financial services. These 'banking deserts' are primarily poorer areas, where individuals do not tend to have a place to deposit paychecks or save for retirement. Instead, they can be preyed upon by predatory lending businesses, like check-cashing/payday loan centers that charge exorbitant fees. This model could potentially make the case for increasing investment in underserved locations, if it demonstrates that improved access to financial institutions predicts a higher Z-score and better banking stability.

Stakeholder Pitch

To the relevant stakeholders at a financial institution, I would propose that they use this model to test how the level of financial access they provide indicates the stability of their financial ecosystem. They can determine which areas of their footprint are indicating a higher risk for instability and use that knowledge to solidify their network of financial institutions.

Data Source

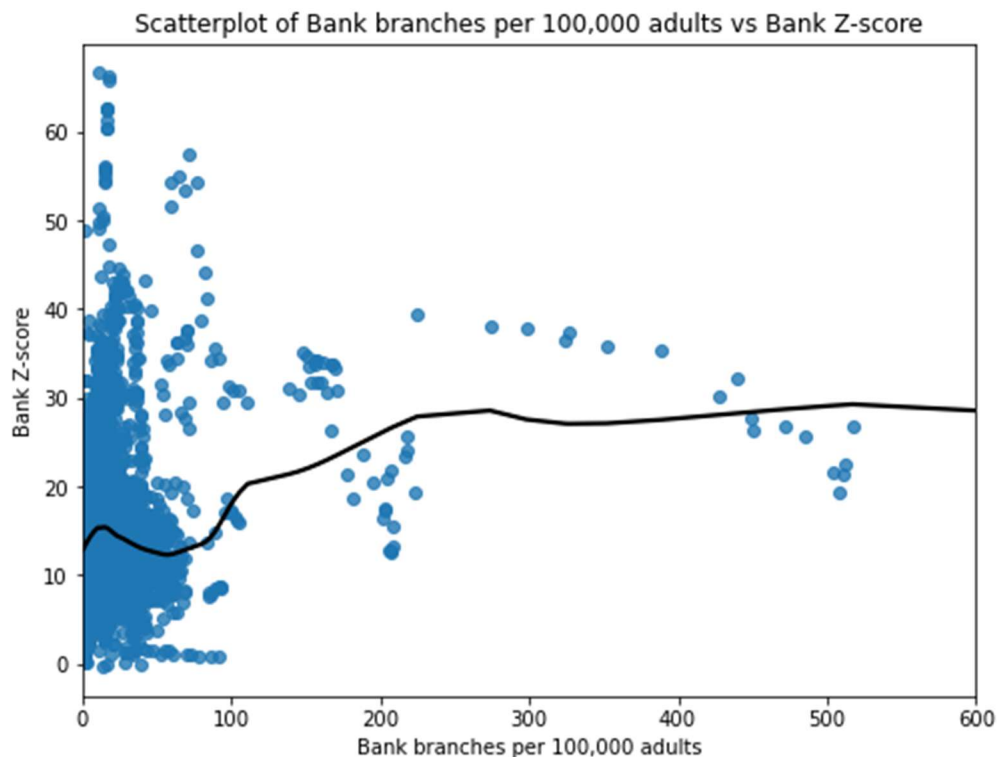
The World Bank Group publishes the Z-Scores of over 200 countries, over about 20 years. They also publish other studies about the economies, giving “World Development Indicator statistics, that give a deeper picture of a country’s overall financial system, beyond the accounting data that drives the Z-Score.

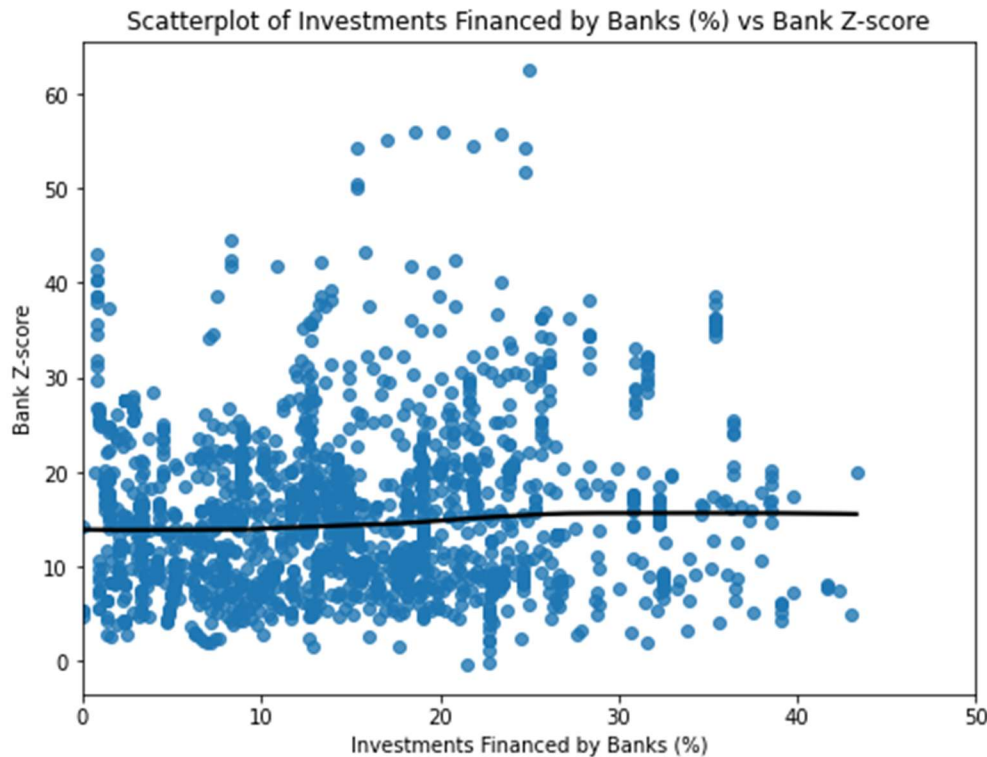
Particularly, there is a subset category called “Financial Institutions: Access” (Series Code "GFDD.AI") that measure how much access a country has to financial products. With this data, we can build a model that attempts to predict the financial stability of a banking system based on how much access the people of the community have to financial institutions.

Data Source: <https://databank.worldbank.org/source/world-development-indicators#>

EDA

In the beginning of the project, we explored a wide variety of features in the World Bank set of development indicators. After some preliminary scatterplots helped us settle on the “Financial Institutions: Access” set of indicators, we checked scatterplots for some of the indicators to see if there appeared to be any relationship between Bank Z-Score and the indicators.





Based on the visual relationships in the scatterplots, the “Financial Institutions: Access” set of indicators looked like a promising set of features.

Data Preparation

Steps to prepare the data included:

- Replacing “.” with “NaN”
- Removing years that did not have a Z-Score
- Interpolating values in between years: Surveys for the indicators were not performed every year, so we used linear interpolation to fill in null values between the survey values.
- We backfilled data to remove some of the “NaN” that interpolation didn’t catch.
- Pivoting: The original data had one row per indicator, per country, with years as columns. We wanted one row per country per year, with each indicator as a column. So a series of two pivots was required.
- Dropped rows that had either no Z-Score, or no features.
- In Milestone 4, we came back and corrected the timing of our backfill, so that values for one country didn’t accidentally backfill into another country.
- In Milestones 3 and 4, we experimented with KNN Imputation vs filling NaN with a mean value. Ultimately, the corrected backfilling, combined with filling with the mean, ended up being the best combination.
- We created dummy features which called out which values were NaN for each feature, before the NaN value was replaced with the mean. The idea was that not having a value might be important information for the model.

Model Building

The model selected was the Random Forest Model. This was chosen because tree models can use the presence of null values as a possible branch. In Milestone 3, Random Forest Models were built with both KNN Imputation and Backfilling. Backfilling proved to have the best results.

Evaluation

At the end of Milestone 3, we had a model with a test R-squared of .926, which seemed a little too good to be true. After reflecting on the nature of the data, we realized that by having one line in the data per country per year, it was possible that we had very similar rows in both the training and the test data. “France 2006” was likely to be very similar to “France 2007” so if one was in test and the other in training, then it would be very easy for the model to predict an accurate Z-Score for France.

To correct this, in Milestone 4, we created a new test set, using an alphabetical split of the countries. Countries with codes from “SGP” and later would be in the test group only, and the model would be seeing countries it hadn’t seen before when the test was completed.

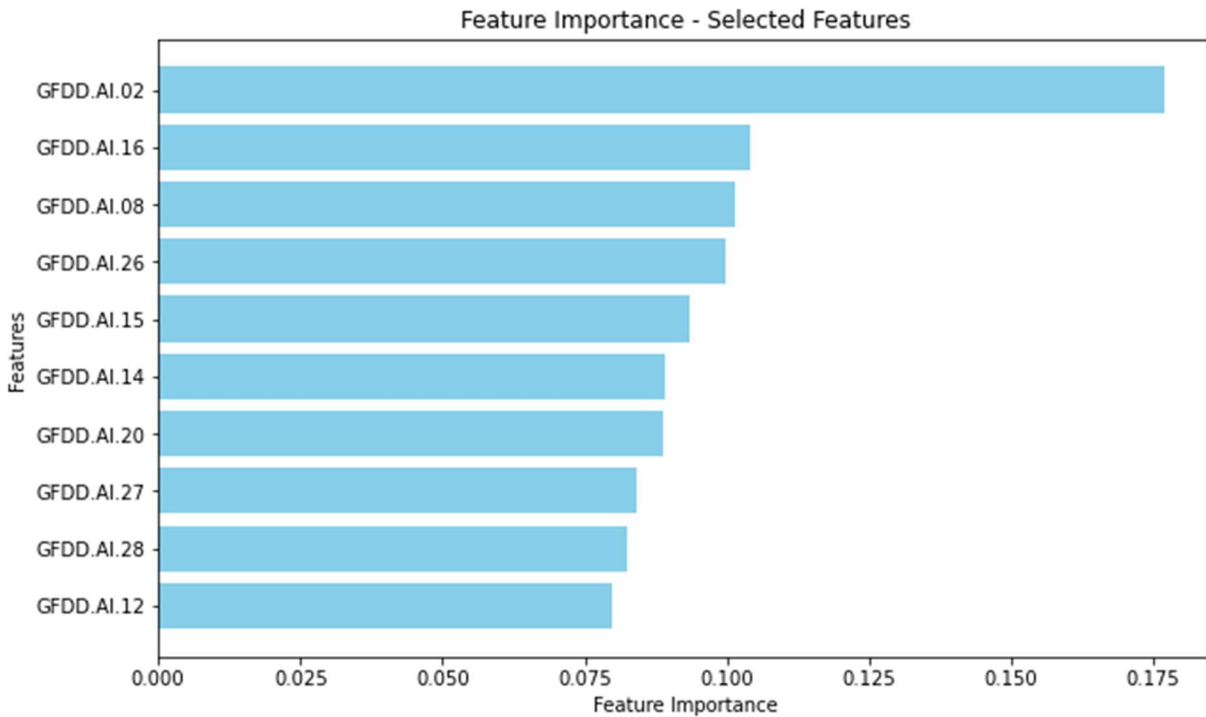
With this correction, the Test R-squared fell to .016, meaning that the model was barely outperforming the mean when it was given countries it had never seen before. We used Recursive Feature Elimination to eliminate as many features as possible.

This method improved our Test set R-Squared to almost 15%. By using the 10 best features relating to Financial Institution: Access, we were able to create a model that explained about 15% of the variability in Z-Scores among countries.

Interpretation

The best version of the model was able to explain 15% of the variability of Z-Score in the test data set. The Z-Score is an accounting metric, and based on these findings, it paints a picture of a financial environment that is too complex to be modeled by information on financial services access only. But we now have the 10 best features, which could be included with future research to create a larger, more comprehensive model of a country’s banking system.

Of those 10 features, AI.02 made the highest contribution to the model, with about an 18% share. The rest contributed between 8-11%. AI.02 corresponds with “Bank Branches per 100,00 adults” which indicates that brick-and-mortar buildings is the largest contributor to banking stability in a country, out of all the “financial institution access” features. However, features AI.14, AI.15, and AI.16 are all on the list as well. Each of these track access to various types of loans, and combine to contribute to about 25% of the model. AI.15 tracks loans from private lenders, and AI.16, the #2 most important feature, tracks loans from employers. This indicates that access to loans is also an important factor in banking stability, even when the loans don’t come from a traditional brick and mortar building.



Series Code	Series Name
GFDD.AI.02	Bank branches per 100,000 adults
GFDD.AI.08	Account used for business purposes (% age 15+)
GFDD.AI.12	Saved any money in the past year (% age 15+)
GFDD.AI.14	Loan in the past year (% age 15+)
GFDD.AI.15	Loan from a private lender in the past year (% age 15+)
GFDD.AI.16	Loan from an employer in the past year (% age 15+)
GFDD.AI.20	Credit card (% age 15+)
GFDD.AI.26	Depositing/withdrawing at least once in a typical month (% age 15+)
GFDD.AI.27	Firms with a checking or savings account (%)
GFDD.AI.28	Firms using banks to finance investments (%)

Overall, this project demonstrated that access to financial institutions contributes to the financial stability Z-Score of a country's banking system, but that it does not paint the full picture.

Steps for Deployment

This model would require modifications before being ready for deployment. At this stage it cannot be used in isolation to make a strong prediction of Z-Score, but it could be used to give an estimate of how much each feature contributes to the Z-Score. For example, an interface could be created using this model to estimate the impact on Z-Score from adding 100 branches in a country. An application like that would be potentially useful for a bank to help plan a footprint expansion strategy that could best increase stability. So despite the fact that this model does not fully predict the Z-Score, it can still fulfill the Stakeholder Pitch, and help an organization plan its footprint.

Recommendations

Ultimately, this model should be considered as a piece of a larger suite of models. I would recommend repeating this analysis with other categories of development indicators. There were other financial-related indicators besides access, such as indicators tracking assets, interest rates and exchange rates. There were also indicators tracking the economic policy and debt of a country (World Bank Group, “World Development Indicators”).

This work could be used as a blueprint to find the top 10 features for each of these categories, creating a catalogue of the most important features across many categories of development indicators. Ultimately, this could help us reach our goal of predicting a country’s financial stability through developmental indicators.

Challenges and Opportunities

The continued challenge for this data set will be finding the best way to handle the sporadic timing of the years in which the surveys are conducted. As the top features are discovered, it might be worth returning to the data prep stage and finding interpolation strategies that work best for those specific features. Although backfilling ended up being the strategy that helped us arrive at our list of 10 features, it may be that KNN Imputation might be a good strategy for improving the performance of these 10 top features.

The greatest opportunity I see in this project is that we may be able to create a model that gives a more robust picture of what contributes to banking system stability than the single accounting metric that Z-Score provides. At the top of this project we noted that Z-Score’s limitation is that it attempts to describe a country’s financial stability through just a handful of accounting data points. This research project could potentially bring in a wide selection of high-performing features that predict that same metric, and in doing so, could provide a much more holistic look at what contributes to a healthy financial banking system.

References

World Bank Group. (n.d.). *Financial Stability*. World Bank Data Bank. Retrieved February 28, 2025, from <https://www.worldbank.org/en/publication/gfdr/gfdr-2016/background/financial-stability>

World Bank Group. (n.d.). *Metadata Glossary*. World Bank Data Bank. Retrieved February 28, 2025, from <https://databank.worldbank.org/metadataglossary/global-financial-development/series/GFDD.SI.01>

World Bank Group. (n.d.). *World Development Indicators*. World Bank Data Bank. Retrieved March 2, 2025, from <https://databank.worldbank.org/source/world-development-indicators#>